

1.1 Добавление формы для создания новых опросов

Цель:

Добавить веб-страницу с формой, через которую пользователь может:

- ввести вопрос (question_text)
- ввести варианты ответов (каждый на новой строке)
- после отправки формы - автоматически создать объект Question и связанные объекты Choice

Шаги реализации:

1. Создание формы

```
forms.py ×
polls > forms.py > QuestionForm
1  from django import forms
2  from .models import Question
3
4  class QuestionForm(forms.Form):
5      question_text = forms.CharField(
6          label='Question text',
7          max_length=200,
8          widget=forms.TextInput(attrs={'class': 'form-control'})
9      )
10     choices = forms.CharField(
11         label='Answer options (one per line)',
12         widget=forms.Textarea(attrs={'class': 'form-control', 'rows': 5})
13     )
```

2. Создание представления view

```
from django.contrib import messages
from .forms import QuestionForm
```

```
views.py ×
polls > views.py > create_poll
55 def create_poll(request):
56     if request.method == 'POST':
57         form = QuestionForm(request.POST)
58         if form.is_valid():
59             question_text = form.cleaned_data['question_text']
60             choices_text = form.cleaned_data['choices'].split('\n')
61
62             question = Question.objects.create(question_text=question_text, pub_date=timezone.now())
63             for choice in choices_text:
64                 choice = choice.strip()
65                 if choice:
66                     Choice.objects.create(question=question, choice_text=choice)
67
68             messages.success(request, 'Poll created successfully!')
69             return HttpResponseRedirect(reverse('polls:index'))
70     else:
71         form = QuestionForm()
72     return render(request, 'polls/create_poll.html', {'form': form})
```

3. Добавление шаблона base.html

```
base.html x
polls > templates > polls > base.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <title>Voting app</title>
6     <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">
7 </head>
8 <body class="container mt-4">
9     <h1 class="mb-4"><a href="{% url 'polls:index' %}" style="text-decoration: none; color: inherit;">Polls</a></h1>
10
11     {% if messages %}
12         {% for message in messages %}
13             <div class="alert alert-info">{{ message }}</div>
14         {% endfor %}
15     {% endif %}
16
17     {% block content %}
18     {% endblock %}
19 </body>
20 </html>
```

4. Добавление шаблона

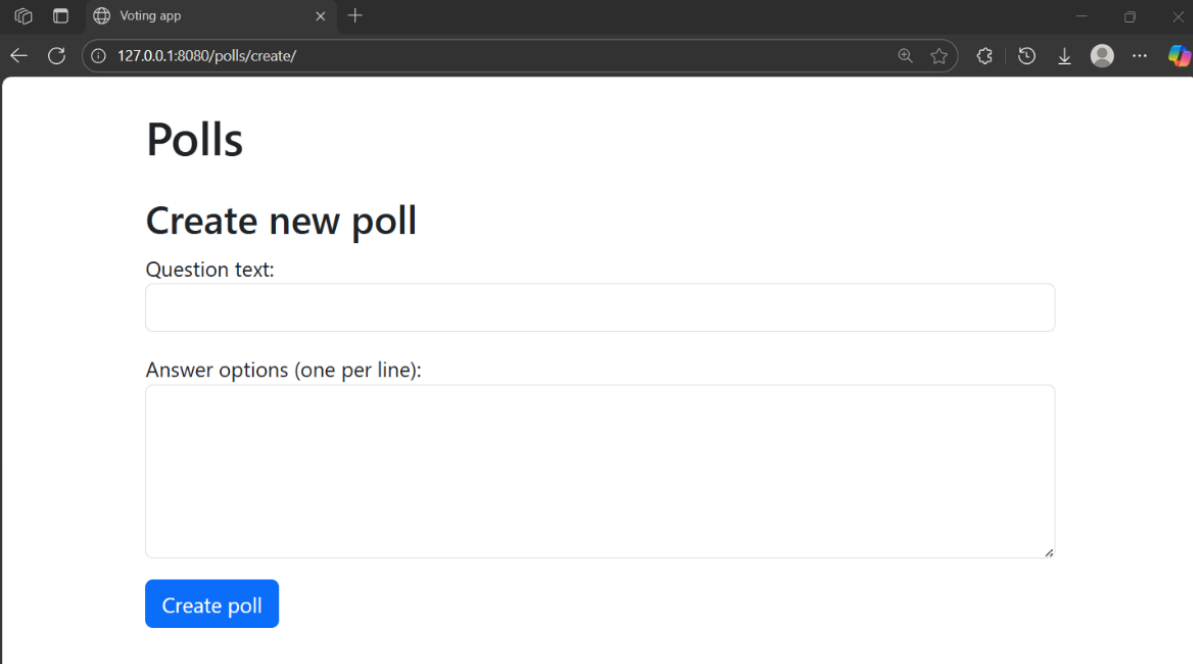
```
create_poll.html x
polls > templates > polls > create_poll.html > ...
1 {% extends 'polls/base.html' %}
2
3 {% block content %}
4 <h2>Create new poll</h2>
5 <form method="post">
6     {% csrf_token %}
7     {{ form.as_p }}
8     <button type="submit" class="btn btn-primary">Create poll</button>
9 </form>
10 {% endblock %}
```

5. Регистрация маршрута. Добавление create/

```
urls.py x
polls > urls.py > ...
1 from django.urls import path
2
3 from . import views
4
5 app_name = "polls"
6 urlpatterns = [
7     path("", views.IndexView.as_view(), name="index"),
8     path("create/", views.create_poll, name="create_poll"),
9     path("<int:pk>", views.DetailView.as_view(), name="detail"),
10    path("<int:pk>/results/", views.ResultsView.as_view(), name="results"),
11    path("<int:question_id>/vote/", views.vote, name="vote"),
12 ]
```

6. Результат:

Теперь по адресу <http://127.0.0.1:8080/polls/create/> можно создавать новые опросы.



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8080/polls/create/`. The page content includes a heading **Polls**, a sub-heading **Create new poll**, a label **Question text:** followed by a text input field, a label **Answer options (one per line):** followed by a larger text area, and a blue button labeled **Create poll** at the bottom.

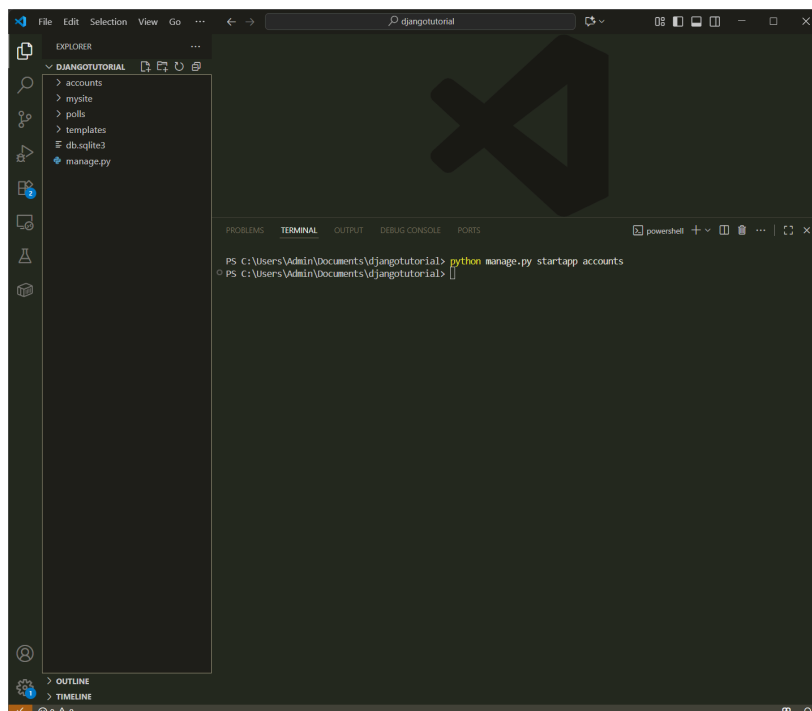
1.2. Аутентификация и регистрация пользователей

Цель:

Добавить систему входа, выхода и регистрации пользователей.

Шаги реализации:

1. Создание нового приложения



Добавление 'accounts' в INSTALLED_APPS

```
settings.py X
mysite > settings.py > ...
30
31 # Application definition
32
33 INSTALLED_APPS = [
34     'polls.apps.PollsConfig',
35     'accounts',
36     'django.contrib.admin',
37     'django.contrib.auth',
38     'django.contrib.contenttypes',
39     'django.contrib.sessions',
40     'django.contrib.messages',
41     'django.contrib.staticfiles',
42 ]
```

2. Настройка формы регистрации и входа

```
forms.py X
accounts > forms.py > RegisterForm > Meta
1 from django import forms
2 from django.contrib.auth.models import User
3 from django.contrib.auth.forms import UserCreationForm
4
5 class RegisterForm(UserCreationForm):
6     email = forms.EmailField(required=True)
7
8     class Meta:
9         model = User
10        fields = ['username', 'email', 'password1', 'password2']
```

3. Добавление views

```
views.py X
accounts > views.py > logout_view
1 from django.shortcuts import render, redirect
2 from django.contrib.auth import authenticate, login, logout
3 from django.contrib import messages
4 from .forms import RegisterForm
5
6 def register_view(request):
7     if request.method == 'POST':
8         form = RegisterForm(request.POST)
9         if form.is_valid():
10             form.save()
11             messages.success(request, 'Registration successful! You can now log in.')
12             return redirect('login')
13     else:
14         form = RegisterForm()
15         return render(request, 'accounts/register.html', {'form': form})
16
17 def login_view(request):
18     if request.method == 'POST':
19         username = request.POST['username']
20         password = request.POST['password']
21         user = authenticate(request, username=username, password=password)
22         if user is not None:
23             login(request, user)
24             return redirect('polls:index')
25         else:
26             messages.error(request, 'Invalid username or password.')
27         return render(request, 'accounts/login.html')
28
29 def logout_view(request):
30     logout(request)
31     return redirect('polls:index')
```

4. Добавление шаблонов

register:

```
register.html X
accounts > templates > accounts > register.html > form
1 <h2>Registration</h2>
2 <form method="post">
3     {% csrf_token %}
4     {{ form.as_p }}
5     <button type="submit">Register</button>
6 </form>
```

login:

```
login.html X
accounts > templates > accounts > login.html > form
1 <h2>Login</h2>
2 <form method="post">
3     {% csrf_token %}
4     <label>Username:</label>
5     <input type="text" name="username" required><br>
6     <label>Password:</label>
7     <input type="password" name="password" required><br>
8     <button type="submit">Log in</button>
9 </form>
```

5. Настройка маршрутов

в accounts:

```
urls.py X
accounts > urls.py > ...
1 from django.urls import path
2 from . import views
3
4 urlpatterns = [
5     path('register/', views.register_view, name='register'),
6     path('login/', views.login_view, name='login'),
7     path('logout/', views.logout_view, name='logout'),
8 ]
```

в mysite:

```
urls.py X
mysite > urls.py > ...
1 """
2 URL configuration for mysite project.
3
4 The 'urlpatterns' list routes URLs to views. For more information please see:
5     https://docs.djangoproject.com/en/5.2/topics/http/urls/
6 Examples:
7 Function views
8     1. Add an import: from my_app import views
9     2. Add a URL to urlpatterns: path('', views.home, name='home')
10 Class-based views
11     1. Add an import: from other_app.views import Home
12     2. Add a URL to urlpatterns: path('', Home.as_view(), name='home')
13 Including another URLconf
14     1. Import the include() function: from django.urls import include, path
15     2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))
16 """
17 from django.contrib import admin
18 from django.urls import include, path
19
20 urlpatterns = [
21     path("polls/", include("polls.urls")),
22     path("admin/", admin.site.urls),
23     path("accounts/", include("accounts.urls")),
24 ]
25
```

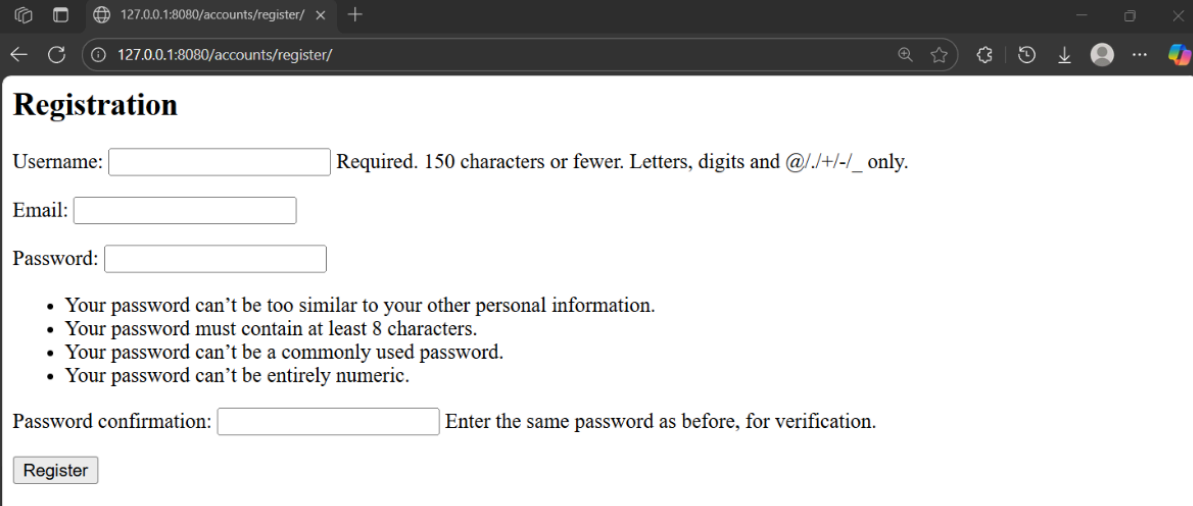
6. Настройка перенаправления

```
settings.py X
mysite > settings.py > ...
121 # Default primary key field type
122 # https://docs.djangoproject.com/en/5.2/ref/settings/#default-auto-field
123
124 DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
125
126 LOGIN_REDIRECT_URL = '/polls/'
127 LOGOUT_REDIRECT_URL = '/polls/'
```

Результат:

Страница регистрации:

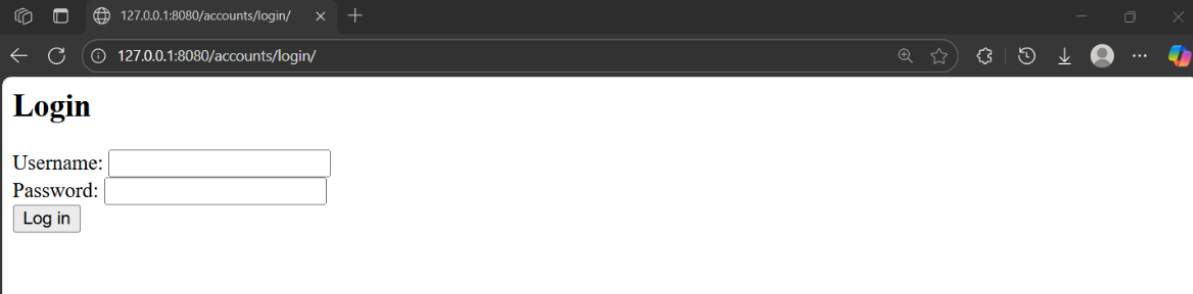
<http://127.0.0.1:8080/accounts/register/>



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8080/accounts/register/`. The page title is "Registration". It contains three input fields: "Username:", "Email:", and "Password:". The "Username:" field has a hint text: "Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only." Below the "Password:" field is a list of password requirements: "Your password can't be too similar to your other personal information.", "Your password must contain at least 8 characters.", "Your password can't be a commonly used password.", and "Your password can't be entirely numeric." Below this list is a "Password confirmation:" field with a hint text: "Enter the same password as before, for verification." At the bottom left is a "Register" button.

Вход:

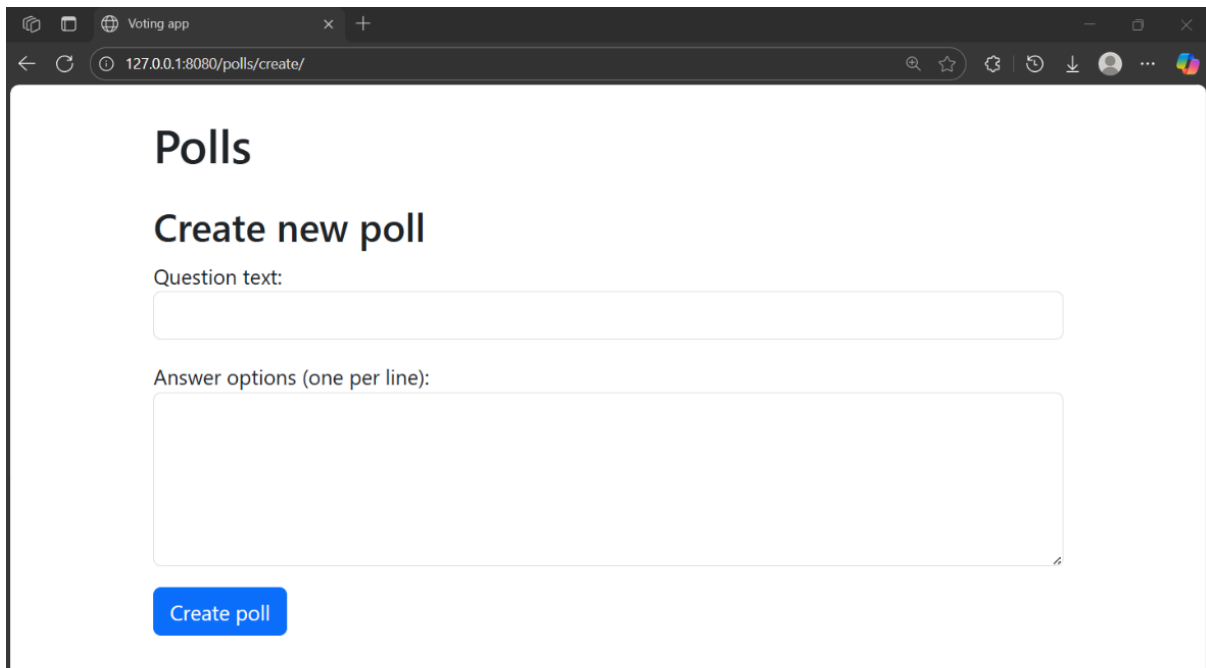
<http://127.0.0.1:8080/accounts/login/>



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8080/accounts/login/`. The page title is "Login". It contains two input fields: "Username:" and "Password:". At the bottom left is a "Log in" button.

Создание опроса:

<http://127.0.0.1:8080/polls/create/>



Polls

Create new poll

Question text:

Answer options (one per line):

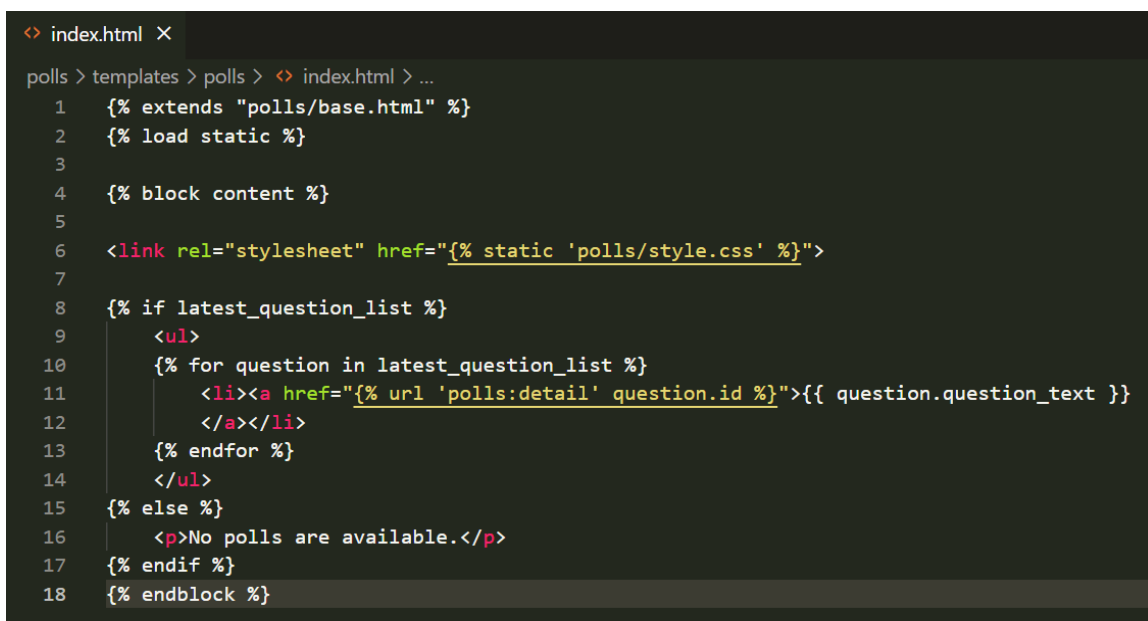
Create poll

7. Использование Django messages для logout



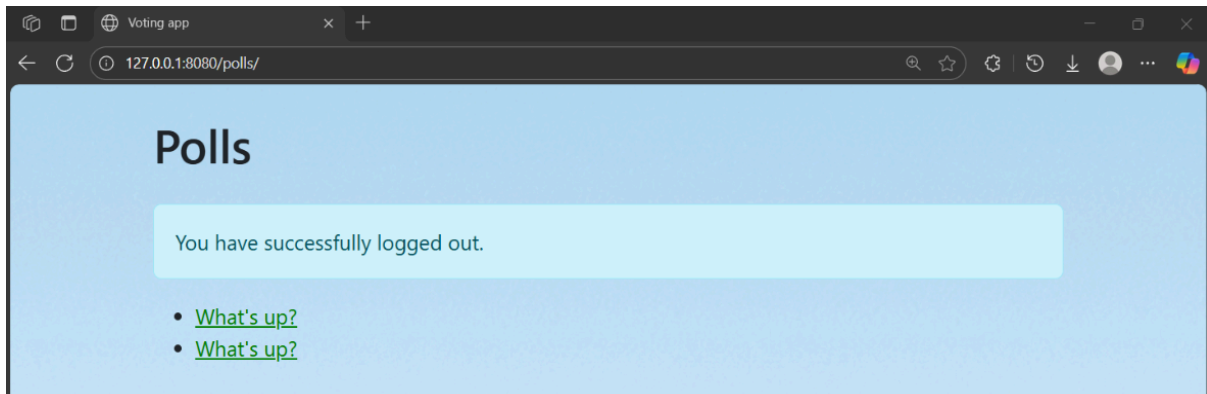
```
views.py X
accounts > views.py > logout_view
29 def logout_view(request):
30     logout(request)
31     messages.info(request, "You have successfully logged out.")
32     return redirect('polls:index')
```

8. Обновление шаблона index

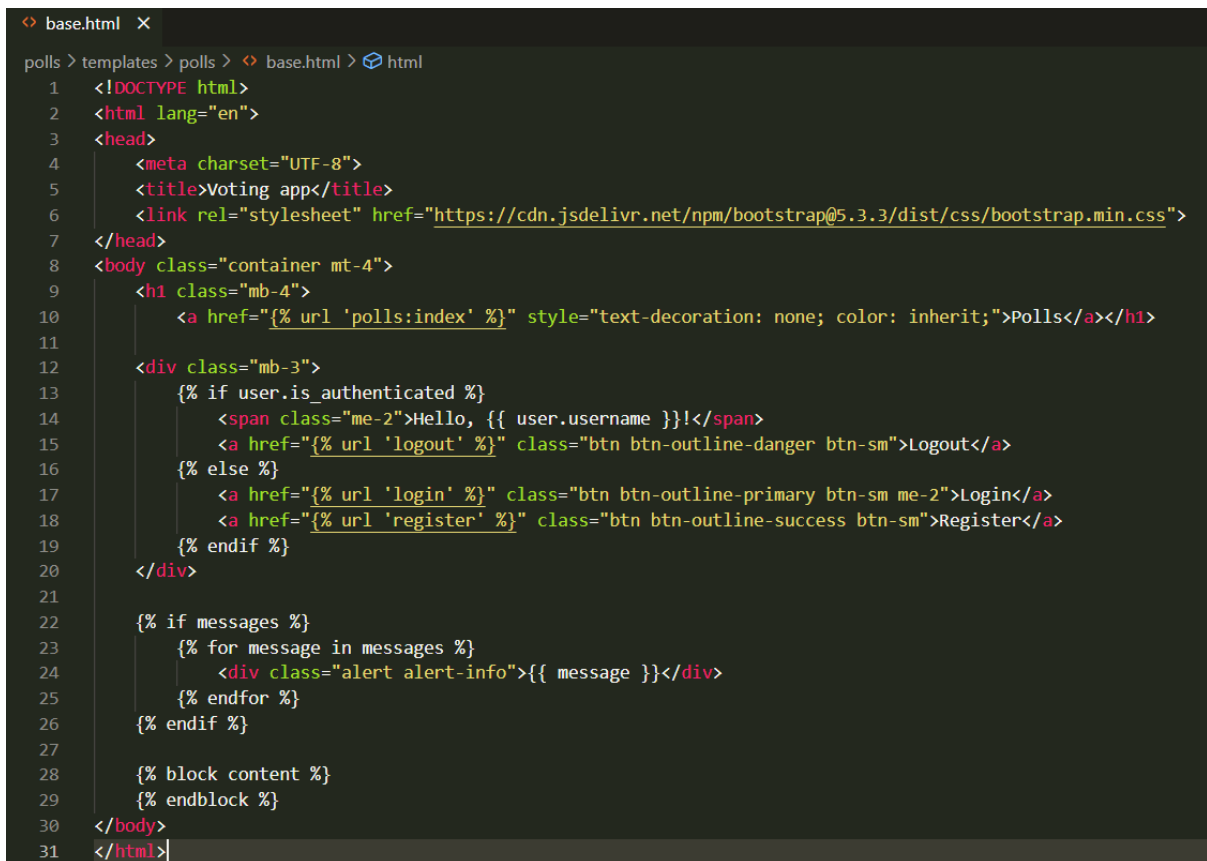


```
index.html X
polls > templates > polls > index.html > ...
1 {% extends "polls/base.html" %}
2 {% load static %}
3
4 {% block content %}
5
6 <link rel="stylesheet" href="{% static 'polls/style.css' %}">
7
8 {% if latest_question_list %}
9     <ul>
10         {% for question in latest_question_list %}
11             <li><a href="{% url 'polls:detail' question.id %}">{{ question.question_text }}
12             </a></li>
13         {% endfor %}
14     </ul>
15 {% else %}
16     <p>No polls are available.</p>
17 {% endif %}
18 {% endblock %}
```

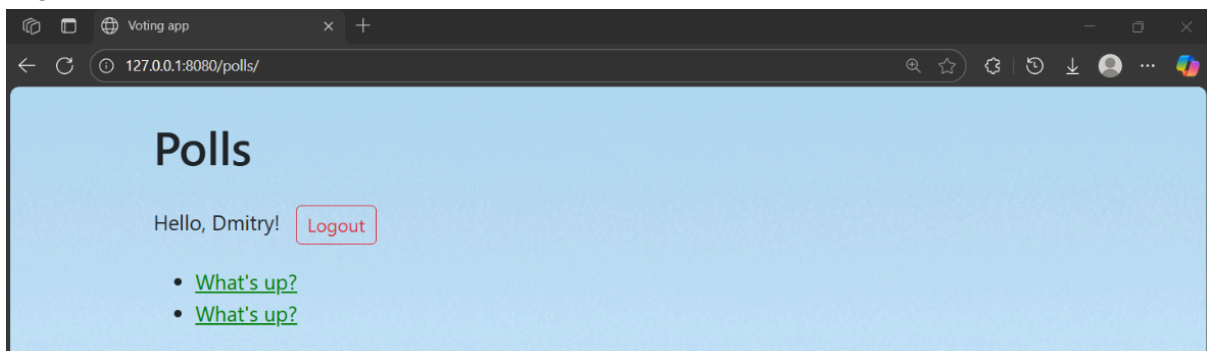
9. Выход



10. Добавление кнопок входа/выхода в base



Login:



Logout:

